

DOCUMENT #7 OF 10 — AI CODING GUARDRAILS

rules.md — AI Developer Guidelines

Strict Coding Standards, State Management Rules, Supabase Best Practices & Error Handling

Target Agent

LLM Developer / AI
Assistant

Enforcement Level

Strict & Zero Regression

Applicable Stack

React 19 + TypeScript +
Supabase

Document Version

v1.0 (Production Rules)

1. TypeScript & Code Quality Rules

RULE 1: Strict TypeScript Typing (No 'any')

Never use the any type in any TypeScript file. All component props, Supabase responses, API payloads, and state objects must have explicit interfaces or types defined in src/types/.

RULE 2: Component Modularity & Clean File Structure

Keep components small and single-purpose (≤ 200 lines). Extract complex logic into custom React hooks (e.g., useResumeParser, useSpeechToText, useCandidateRanking).

2. UI/UX & Glassmorphism Styling Rules

RULE 3: Pure Vanilla CSS Design Tokens

Do not install heavy third-party UI libraries unless strictly instructed. Leverage custom CSS variables for themes (--bg-glass, --border-glass, --accent-primary) with smooth transitions.

RULE 4: Mandatory Interactive States

Every interactive element (buttons, upload cards, filters) must implement: :hover, :focus-visible, :active, and :disabled states with accessible aria labels.

3. Supabase & Backend Communication Rules

RULE 5: Client-Side Security & Key Protection

Never hardcode API keys or service role secrets in client-side code. Use import.meta.env.VITE_SUPABASE_URL and import.meta.env.VITE_SUPABASE_ANON_KEY. All privileged LLM calls must pass through authenticated Supabase Edge Functions.

RULE 6: Row-Level Security (RLS) Compliance

Every query to PostgreSQL must respect the authenticated tenant/user context. Always structure write queries to include user_id: auth.uid().

4. Mandatory Runtime & App Integrity Rules

RULE 7: Mandatory Loading States & Skeleton Loaders

Every asynchronous operation (file parsing, candidate leaderboard fetch, interview streaming) must display a clear skeleton placeholder or spinner to prevent layout shifts.

RULE 8: Robust Error Boundaries & User Feedback

Wrap critical view trees in React Error Boundaries. Always display user-friendly toast notifications for network errors, invalid file formats, or API rate limits.

RULE 9: Client-Side File Validation

Validate resumes before sending them to storage: enforce allowed MIME types (`application/pdf`, `application/vnd.openxmlformats-officedocument.wordprocessingml.document`) and max file size ($\leq 5\text{MB}$).

5. AI & Speech-to-Text Integration Standards

RULE 10: Graceful Fallbacks for Browser Speech APIs

Check for browser compatibility before initiating `window.SpeechRecognition` or `window.webkitSpeechRecognition`. If unsupported, seamlessly provide a standard text-area input fallback.